Unit 19: Data Structures & Algorithms

Authorized Assignment Brief

Unit Number and Title	Unit 19: Data Structures & Algorithms
Academic Year	2025 -2026 – Spring Semester (01/2026 – 05/2026)
Unit Tutor	MSc. Luong Tran Ngoc Khiet
Assignment Title	Final Assignment - Individual
Issue Date	**17 Apr 2026**
Submission Date	**09 May 2026**

Submission Format

- Must include an approved cover sheet provided by Saigon Business school.
- Students must sign the declaration of original work in the cover sheet.
- Individual project
- Format: Present on template report, write your full names and fill in all task.
- It is required to clearly state the formula used or explain how to do it
- Submit your assignment in digital format (PDF, Ms. Word document) to Google Classroom.

Unit Learning Outcomes

- **LO1**: Examine abstract data types, concrete data structures and algorithms
- **LO2:** Specify abstract data types and algorithms in a formal notation
- **LO3:** Implement complex data structures and algorithms
- **LO4:** Assess the effectiveness of data structures and algorithms...

Transferable skills and competencies developed

Critical Thinking and Analysis: Evaluating the complexity and efficiency of various data structures and algorithms

Problem-Solving: Designing and implementing solutions for complex computer problems such as network analysis and process control.

Reasoning and Interpretation: Understanding the trade-offs between different abstract data types (ADTs) and concrete implementations.

Technical Communication: Using formal notation and specifications to describe software behavior.

Synthesis: Integrating abstract specifications with concrete programming paradigms to create maintainable code

Vocational scenario: HealthData Insights

Context:

You are a Software Engineer for "Swift-Load Logistics." The company needs a system to manage goods loaded onto delivery trucks. Each item has a Name, Type, and Weight. Your task is to develop a suite of tools to sort these items, optimize the truck's capacity, and manage the delivery route efficiently.

Assignment activity and guidance

Project Phases & Specific Tasks

Task 1: System Architecture and Cargo Sorting
This task focuses on the initial design of the logistics platform, ensuring data integrity and basic organizational logic.

- **Design Specification (P1):** Create a design specification for the "Goods" data structure (containing Name, Type, and Weight). You must explain the valid operations that can be carried out on this structure, such as initialization and weight updates.
- **Queue Illustration (M1):** Illustrate with a concrete example how a **First In First Out (FIFO)** queue structure is used to manage the sequence of trucks entering the Swift-Load loading bay.
- **Sorting Comparison (M2):** Take a sample of 12 cargo weights and sort them in descending order using a basic algorithm (e.g., Bubble Sort). Compare the performance of this method against a more advanced sorting algorithm like QuickSort.
- **Data Encapsulation (M3):** Examine the advantages of using **encapsulation and information hiding** when designing the "Goods" ADT to ensure that cargo metadata remains secure within the system.
- **OOP Foundations (D2):** Discuss the view that the **imperative ADTs** (like your Goods structure) serve as a fundamental basis for **Object-Orientation**, offering a justification based on your system's design.

Task 2: Operational Optimization and Efficiency
This task moves into the theoretical and analytical aspects of resource management and routing.

- **Stack Operations for Process Control (P2):** Determine the operations of a **memory stack** and describe how it is used to implement function calls within the truck-loading software.
- **Formal Specification (P3):** Provide a formal specification for a **software stack** (used for LIFO loading) using an **imperative definition**.
- **Routing Analysis (D1):** Using illustrations of your distribution map, **analyse the operation** of two network shortest path algorithms (e.g., **Dijkstra's and A***) to determine the most efficient delivery routes.
- **Effectiveness via Big O (P6):** Discuss how **asymptotic analysis (Big O)** can be used to assess the effectiveness of your routing and loading algorithms.
- **Measuring Efficiency (P7):** Determine and explain **two ways** in which the efficiency of your algorithms can be measured (e.g., execution time vs. memory usage), providing examples from your code.
- **Trade-off Interpretation (M5):** Interpret what a **trade-off** is when specifying an ADT for the logistics system (e.g., choosing a faster but more memory-intensive structure), using a specific example to support your answer.

Task 3: Implementation and Technical Critique

This final task involves the hands-on development of a complex system and a critical review of its performance.

- **Complex Implementation (P4 & M4):** Implement a **complex ADT and algorithm** (specifically, an **AVL Tree**) in an executable programming language to manage warehouse inventory. You must **demonstrate** how this implementation successfully solves the well-defined problem of high-speed data retrieval for your logistics system.
- **Robustness and Testing (P5):** Implement robust **error handling** within your code (e.g., managing null cargo entries) and provide a report of your test results.
- **Critical Evaluation (D3):** Provide a **critical evaluation of the complexity** of your implemented AVL tree compared to a simple linear search algorithm using Big O notation.
- **Implementation Independence (D4):** Evaluate **three benefits** of using **implementation-independent data structures** to ensure that the Swift-Load software can be easily maintained or ported to other platforms in the future

Deliverables:

- Assignment report: Reading [NOTICE ON COVER SHEET & ASSIGNMENT SUBMISSION] in classroom.
- Share link **github.com**: Store the C++ source code on github.com and share it with anyone who has the link. Include this link in the appendix.

Recommended resources

Please note that the resources listed are examples for you to use as a starting point in your research – the list is not definitive.

- Course lectures
- Q&A Stackoverflow . Available at: https://stackoverflow.com/

Learning Outcomes and Assessment Criteria

Pass	Merit	Distinction
LO1 Examine abstract data types, concrete data structures and algorithms		
P1 Create a design specification for data structures, explaining the valid operations that can be carried out on the structures. **P2** Determine the operations of a memory stack and how it is used to implement function calls in a computer.	**M1** Illustrate, with an example, a concrete data structure for a First in First out (FIFO) queue. **M2** Compare the performance of two sorting algorithms.	**D1** Analyse the operation, using illustrations, of two network shortest path algorithms, providing an example of each.
LO2 Specify abstract data types and algorithms in a formal notation		
P3 Specify the abstract data type for a software stack using an imperative definition.	**M3** Examine the advantages of encapsulation and information hiding when using an ADT.	**D2** Discuss the view that imperative ADTs are a basis for object orientation offering a justification for the view.

Pass	Merit	Distinction
LO3 Implement complex data structures and algorithms		
P4 Implement a complex ADT and algorithm in an executable programming language to solve a well-defined problem. **P5** Implement error handling and report test results.	**M4** Demonstrate how the implementation of an ADT/algorithm solves a well-defined problem.	**D3** Critically evaluate the complexity of an implemented ADT/algorithm.
LO4 Assess the effectiveness of data structures and algorithms		
P6 Discuss how asymptotic analysis can be used to assess the effectiveness of an algorithm. **P7** Determine two ways in which the efficiency of an algorithm can be measured, illustrating your answer with an example.	**M5** Interpret what a trade-off is when specifying an ADT, using an example to support your answer.	**D4** Evaluate three benefits of using implementation independent data structures.